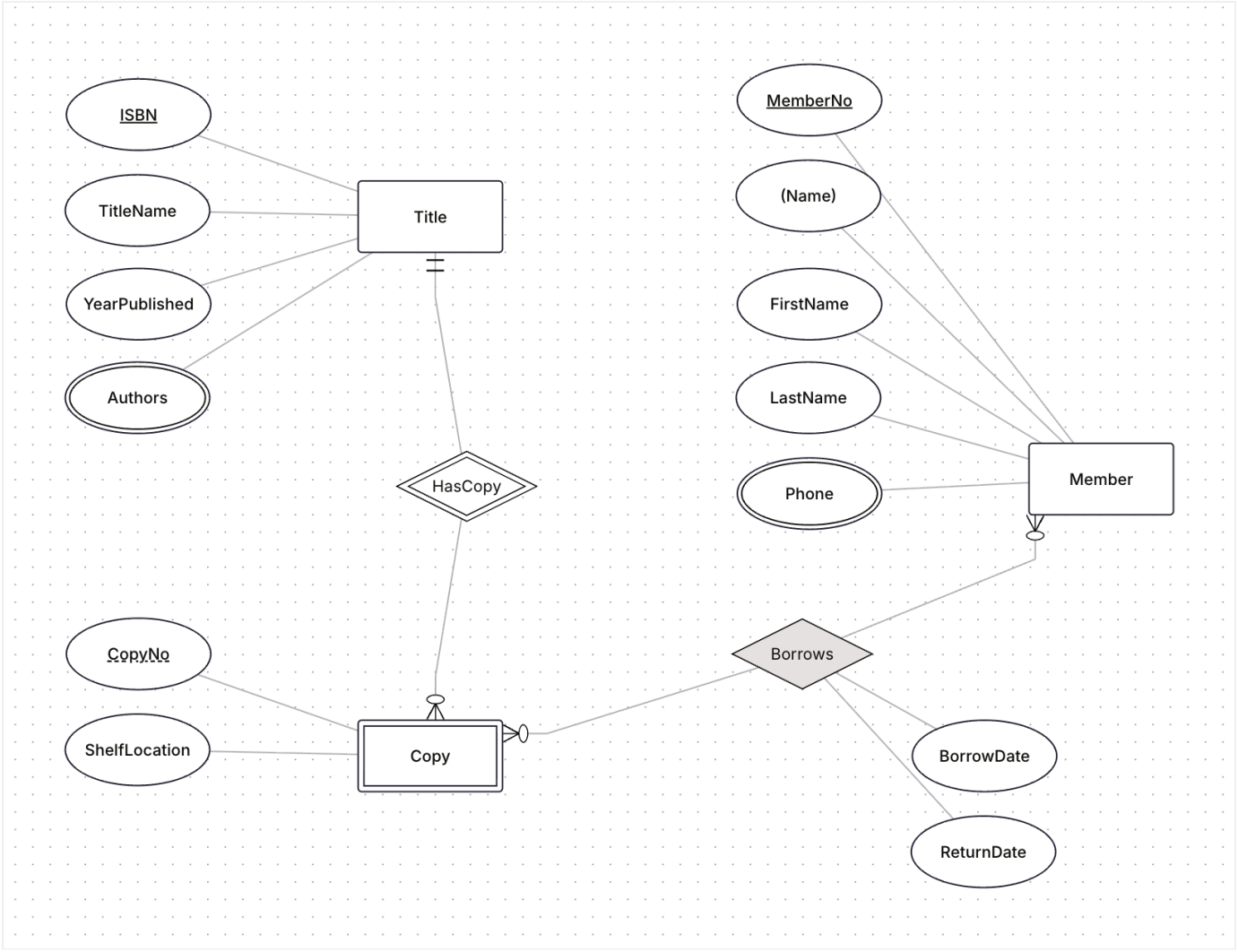
IT079IU — Lab 1: Introduction to the Entity–Relationship Diagram

International University – VNU-HCM · School of Computer Science and Engineering · IT079IU Principles of Database Management

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Lab number: Lab 1

Part B — A small library (4 marks)



Chen notation drawn in ERDPlus. Keys are underlined; a dashed underline marks a partial key; a double rectangle marks a weak entity; a double diamond marks an identifying relationship; a double ellipse marks a multivalued attribute; a dashed ellipse marks a derived attribute.

> **Entities:** Title, Copy (weak), Member.
>
> **Relationships:** Title – HasCopy – Copy is 1:N, total on the Copy side, because a copy cannot exist without its title. Member – Borrows – Copy is M:N with the attributes borrow date and return date, partial on both sides, because a member may borrow nothing and a copy may never be borrowed.
>
> **Key of the weak entity:** ISBN + copy number.

Why Copy is a weak entity

The specification says a copy is numbered 1, 2, 3 … *within its title*, so copy number 2 means nothing unless the title is known. Copy therefore has no identifier of its own: `CopyNo` is only a partial key (dashed underline) and the copy is

identified by the pair ISBN + CopyNo through the identifying relationship **HasCopy** (double diamond, total participation on the Copy side).

Why borrow date and return date sit on the diamond

They describe a borrowing *event*, not a member and not a copy: the same member borrows the same copy several times over time, each time with a different pair of dates. A fact about a pair belongs on the relationship, so `BorrowDate` and `ReturnDate` are attributes of **Borrows**, not of Member or Copy.

Why the authors are an attribute and not an entity

The specification gives authors nothing but a name — no code, no address, no relationship of their own — so by the test in the lab sheet ("does it have properties of its own beyond a name?") Authors is a multivalued attribute of Title, drawn as a double ellipse.

Part C — A clinic (6 marks)

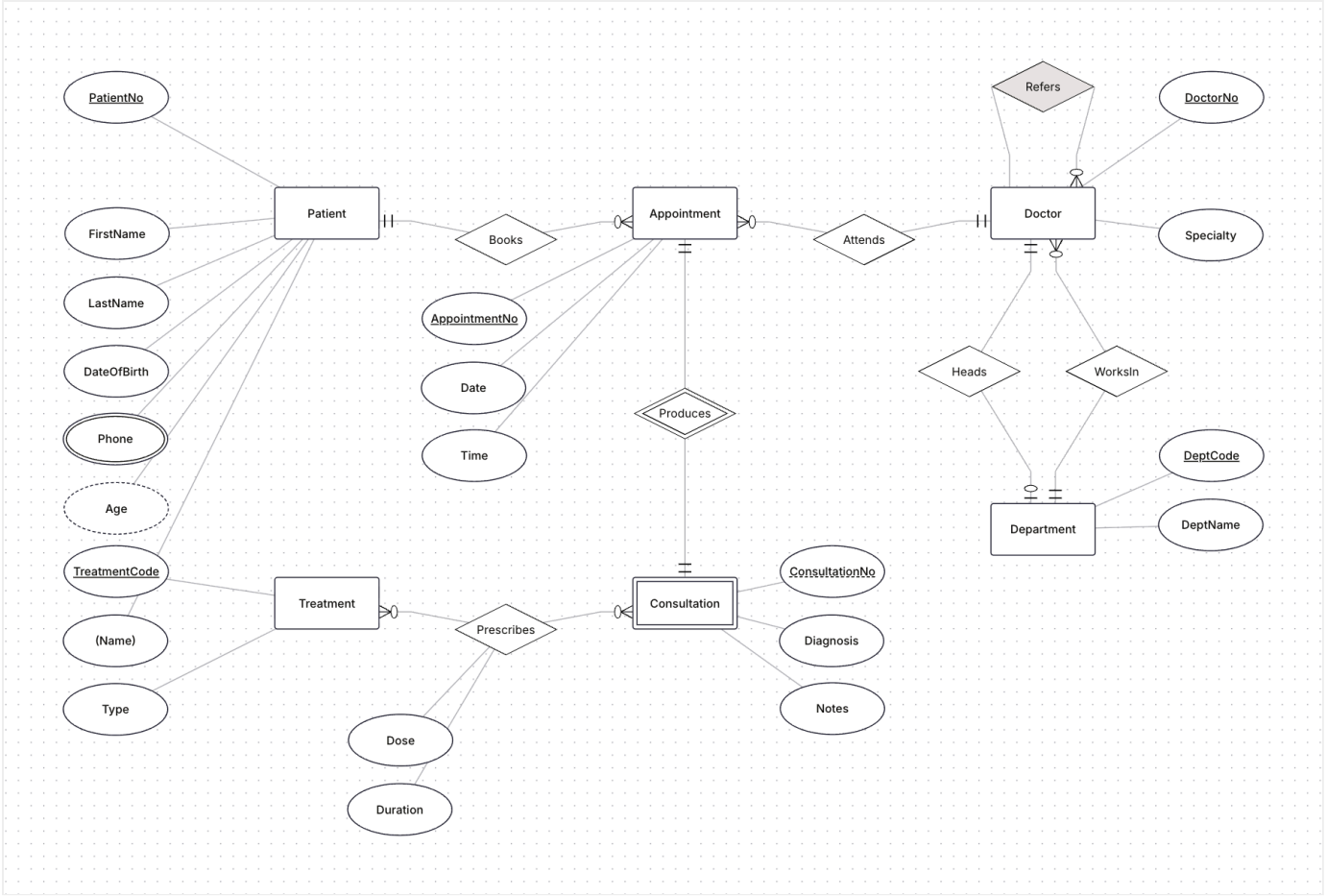


Table of relationships

Relationship	Entities joined	Ratio	Participation	Sentence from the specification that justifies it
Books	Patient – Appointment	1 : N	Optional on Patient, Mandatory on Appointment	"A patient makes an appointment with one doctor for a date and a time." + "A patient may have many appointments over time." One appointment has exactly one patient (Mandatory, One); a patient may have none yet (Optional, Many).
Attends	Appointment – Doctor	1 : N	Optional on Doctor, Mandatory on Appointment	"A patient makes an appointment with one doctor" + "a doctor sees many patients." Each appointment belongs to exactly one doctor (Mandatory, One); a doctor may have no appointments (Optional, Many).
Produces (identifying)	Appointment – Consultation (weak)	1 : 1	Mandatory on both sides	"Each appointment produces exactly one consultation record that contains the diagnosis and notes." — so every appointment has exactly one consultation and participation on the Appointment side is total. "A consultation record cannot exist without its appointment." — total participation on the Consultation side as well.
WorksIn	Doctor – Department	N : 1	Mandatory on Doctor, Optional on Department	"Every doctor works in exactly one department." The "exactly one" makes participation on the Doctor side total; "a department has a code, a name" says nothing about it always having a doctor.
Heads	Department – Doctor	1 : 1	Mandatory on Department, Optional on Doctor	"a department ... is headed by exactly one of its own doctors." Each department has exactly one head (Mandatory, One); a doctor does not head a department merely by existing, so the Doctor side is Optional.

Relationship	Entities joined	Ratio	Participation	Sentence from the specification that justifies it
Prescribes	Consultation – Treatment	M : N	Optional on both sides	"A consultation may prescribe several treatments, and the same treatment may be prescribed in many consultations. The dose and the duration are recorded each time a treatment is prescribed."
Refers	Doctor – Doctor (entity with itself)	M : N	Optional on both sides	**Not stated in the specification.** The task sheet requires "at least one relationship of an entity set with itself", but the clinic text contains none, so this relationship was added; see the comment below.

Attributes of relationships

- **Prescribes** carries `Dose` and `Duration`. They are recorded "each time a treatment is prescribed", so they describe the pair (consultation, treatment) rather than either entity on its own.
- **Produces** is the identifying relationship of the weak entity set Consultation, whose partial key is `ConsultationNo`; the full identifier is AppointmentNo + ConsultationNo.

Entity and attribute notes

- **Patient.age** is a *derived* attribute (dashed ellipse): "The clinic displays the patient's age but does not store it."
- **Patient.Phone** is multivalued — "one or more phone numbers" — hence the double ellipse. **Patient.Name** is composite, "a full name made of a first and a last name", hence `FirstName` and `LastName`.
- **Doctor.Specialty** and **Treatment.Type** are plain attributes: the specification gives them only a word, so by the lab's own test they do not become entity sets.
- No entity stores the identifier of another entity, and no junction table appears: Borrows-style many-to-many links are drawn as single diamonds. This is a conceptual model, so no foreign key appears anywhere.

Comment on the hardest design decision

> The hardest decision was **Appointment**. The specification introduces it as an event — "a patient makes an appointment with one doctor for a date and a time" — which suggests a many-to-many link between Patient and Doctor. But the next requirement, that a consultation record "cannot exist without its appointment", needs something that can *own* a weak entity, so Appointment had to be modelled as a **strong entity with its own key, AppointmentNo**, with Books and Attends as two 1:N relationships into it. The obvious alternative, treating Appointment as an associative entity keyed by PatientNo + DoctorNo + Date + Time, was rejected because it would put the identifier of another entity inside an entity, which is exactly the mistake the lab sheet asks us to avoid.
>
> Second, the task sheet requires a relationship of an entity set with itself, but the clinic specification contains none. Rather than leave the requirement unmet, **Refers** was added as Doctor – Refers – Doctor (roles: referring doctor / referred doctor), the most natural clinical referral link. This is disclosed here because it is an addition to the specification, not something the text states.

AI use: An AI assistant was used to derive the entity sets, keys, attributes, relationships, cardinalities and participation constraints from the specification, to draw both diagrams in ERDPlus, and to draft this written description. The design decisions were discussed and approved by me, and I am able to justify every element of both diagrams.